[9] https://sites.google.com/site/murmurhash/

[10] http://www.isthe.com/chongo/tech/comp/fnv/index.html

[11] We could, of course, check our local Bloom filter for the contact, but even when it returns true, without checking the server we have no way to know if it's a false positive!

[12] Here `e` is Euler's number; we'll meet it again in section 4.10.

[13] Bloom filters also are or have been used for a long time in browsers for "safe browsing", basically keeping a black-list of malicious sites. Chromium engine, for instance, replaced it with compressed `PrefixSets` only a few years ago.

[14] "Utilizing Bloom filters for detecting flooding attacks against SIP based services.", Geneiatakis, Dimitris, Nikos Vrakas, and Costas Lambrinoudakis computers & security 28.7 (2009): 578-591.

[15] This section, as well as the following ones, are theory-intensive and feature advanced concepts.

[16] Assuming `m` is the size of the filter's buffer.

[17] `m` *choose* `k` is a binomial coefficient, see https://en.wikipedia.org/wiki/Binomial_coefficient

[18] Any object or value can be serialized to a string (for instance, a binary string…).

[19] This section includes advanced, math-intensive content.

[20] In other words, as we have seen in the previous sections, we assume the hash functions are drawn from a set of universal hash functions.

[21] e^x and `ln(x)` are inverse functions, so that, for `x>0`, $\ln(e^x) = e^{\ln(x)} = x$; that stands true even if `x` is a function (as long as it's always positive).

[22] Chazelle, Bernard; Kilian, Joe; Rubinfeld, Ronitt; Tal, Ayellet (2004), "The Bloomier filter: an efficient data structure for static support lookup tables", Proceedings of the Fifteenth Annual ACM-SIAM Symposium on Discrete Algorithms (PDF), pp. 30–39.

[23] Zhiwang, Cen; Jungang, Xu; Jian, Sun (2010), "A multi-layer Bloom filter for duplicated URL detection", Proc. 3rd International Conference on Advanced Computer Theory and Engineering (ICACTE 2010), 1, pp. V1–586–V1–591, doi:10.1109/ICACTE.2010.5578947

[24] Mitzenmacher, Michael. "Compressed bloom filters." IEEE/ACM Transactions on Networking (TON) 10.5 (2002): 604-612.